CALCULUS FOR AI
Problem Set 02

From Mistakes to Learning

Loss, gradient descent, probability, and matrices

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The big idea

An AI model makes a guess. We measure how wrong it is using a **loss**. The model adjusts its settings to reduce that loss. For classification, it expresses uncertainty with probabilities; inside the model, numbers are organized and transformed using matrices.

Name: _____

Date: _____ **Suggested time:** 75–100 minutes

Total: 100 marks

What you will learn

- calculate and interpret squared-error loss;
- perform a gradient-descent update and explain the learning rate;
- explain why probabilities sum to 1 and why confident mistakes matter;
- calculate a small matrix-times-vector product;
- connect these ideas to the way an AI model learns.

A small guide to the symbols

Symbols you will see

Symbol	Meaning in simple words
y	The correct answer (the truth).
$\hat{y}$	The model's prediction, read "y-hat."
L	Loss: a number measuring how wrong a prediction is. Lower is usually better.
w	A model setting (weight) that training can adjust.
$\frac{dL}{dw}$	The slope of the loss as the weight changes.
η	Learning rate: how large a step the model takes.
p	A probability between 0 and 1.
ln	Natural logarithm; no hand calculation is needed here.
W	A matrix of weights: a rectangular table of numbers.

How to work

- Show your steps; reasoning matters, not only the final number.
- Explain ideas in everyday language when asked.
- Use the formulas provided.

Part A — Measuring a model's mistakes (20 marks)

Big idea: loss turns prediction error into a score we try to reduce. **A1. Squared-error loss** [5 marks]

A model predicts a bus journey's travel time. The true time is 40 minutes.

(a) Prediction is 46 minutes. Find error $\hat{y} - y$ and loss $(\hat{y} - y)^2$. [2]

(b) A second model predicts 37 minutes. Find its loss and say which model is better by this measure. [2]

(c) Why does squaring make both overestimates and underestimates positive? [1]

A2. Mean squared error [5 marks]

House prices are in thousands of dollars. Fill the squared errors and calculate their average (MSE).

House	True price	Prediction	Squared error
A	100	110	_____
B	150	140	_____
C	200	220	_____

A3. Interpret the score [4 marks]

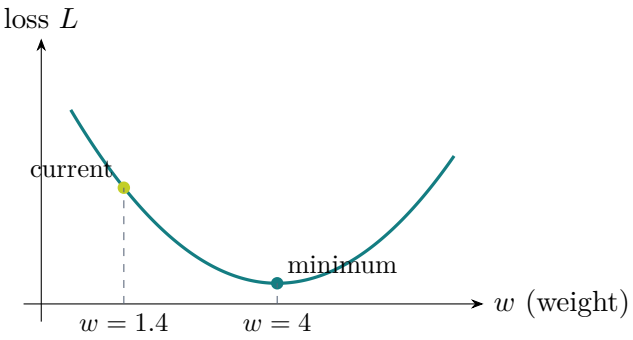
Model A has loss 9 and Model B has loss 25 on the same task and data. Which is doing better? Name one thing this score alone does not tell us.

A4. Choose a loss [6 marks]

A team builds a model predicting whether a student will arrive late. What does low loss mean? Why compare predictions with known outcomes? Why might house-price prediction and spam classification use different loss formulas?

Part B — One step downhill (25 marks)

Big idea: the slope guides how a model adjusts its setting. **B1. Read the loss landscape** [4 marks]

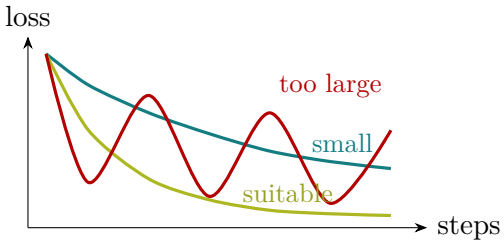


At $w = 1.4$, is the slope negative, zero, or positive? Which direction should the weight move to reduce loss? What does the bottom represent?

B2. Make one update **[8 marks]**

Use $w_{new} = w_{old} - \eta \frac{dL}{dw}$. At $w = 1.5$, slope $= -8$, and learning rate $\eta = 0.2$. Substitute and calculate the new weight. Did it increase or decrease, and why? If the slope were $+8$, what would the new weight be?

B3. Learning rate: size of the stride **[5 marks]**



Identify the too-large learning rate and the slowest curve. Explain why too-large steps can make training unstable.

B4. Follow the numbers **[8 marks]**

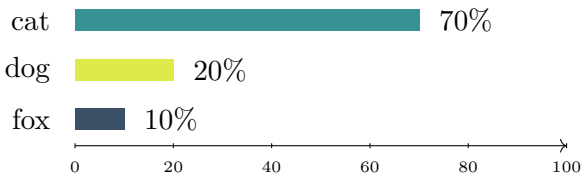
A tiny model predicts $\hat{y} = wx$, with $x = 2$, $y = 8$, loss $L = (\hat{y} - y)^2$, and slope $\frac{dL}{dw} = 2(\hat{y} - y)x$. Initially $w = 1$, $\eta = 0.1$. Find (a) prediction and loss, (b) slope, (c) updated weight, (d) new prediction and loss. Did it improve?

Part C — Probability and confident mistakes (20 marks)

Big idea: probabilities express uncertainty; cross-entropy penalizes giving the correct answer very little probability. **C1. Read the probabilities** [4 marks]



Figure 1: a simple photo of a cat.



Which label is most likely? Verify the total is 100%. Does a 70% cat probability prove the photo is a cat? Explain.

C2. Fix the probabilities [4 marks]

A model assigns 0.55 to cat, 0.35 to dog, and 0.25 to fox. Why is this not a valid probability distribution? If the first two stay fixed, what must the fox probability be?

C3. Which mistake is worse? [6 marks]

A picture really is a dog. Compare: Model A assigns $P(dog) = 0.90$ (loss 0.11), B assigns 0.50 (loss 0.69), C assigns 0.10 (loss 2.30). Which gets smallest and largest loss? Explain why loss grows when probability assigned to the correct answer gets very small.

C4. Evidence changes a belief [6 marks]

Of 100 emails, 20 are spam. Among 20 emails containing "free", 15 are spam. Find (a) the overall spam percentage, (b) the spam percentage among emails containing "free", and (c) explain what this illustrates about evidence.

Part D — Matrices: numbers with structure (20 marks)

Big idea: a matrix is a table. Each row can combine inputs to produce one output. **D1. Read a matrix** [4 marks]

$$X = \begin{bmatrix} 80 & 3 & 10 \\ 120 & 4 & 5 \\ 45 & 1 & 30 \end{bmatrix} \quad \text{columns: area } (\text{m}^2), \text{ rooms, age (years)}$$

How many rows and columns? What does $X_{2,1} = 120$ mean? Which row represents a one-room house? What does a column represent?

D2. Matrix times a vector [8 marks]

$$W = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Calculate Wx using the row rule and show both calculations. Why is this useful in a neural network?

D3. Pixels become numbers [4 marks]

A tiny image uses 0 for white and 9 for dark: $P = \begin{bmatrix} 0 & 9 & 0 \\ 9 & 9 & 9 \\ 0 & 9 & 0 \end{bmatrix}$. Sketch it with dark and white squares. Why can a vision model process this matrix?

D4. Match the shapes [4 marks]
Matrix A has 2 rows and 3 columns. Vector x has 3 entries. How many entries does Ax have? Explain why each row produces one output number.

Part E — Put the story together (15 marks)

E1. The training loop [5 marks]
Number these stages 1–4: update weights; calculate loss; make prediction; calculate slope/gradient (4 marks). Why does the process repeat? (1 mark)

E2. Diagnose training [5 marks]
A model's loss decreases, then jumps up and down by large amounts. Give a likely cause and one sensible thing to try. Explain.

E3. Explain it to a friend [5 marks]
In 4–6 sentences, explain how loss, gradient descent, probability, and matrices fit into one AI system. Include an analogy a beginner would understand.

Optional learning links

These are optional supports, not required reading.

- Google Machine Learning Crash Course — Gradient descent.
- Google — Learning rate and hyperparameters.
- 3Blue1Brown — Gradient descent, visually.
- StatQuest — Cross entropy.
- Google — Multi-class classification and softmax.
- NumPy — Absolute basics.

Final reminder Show your reasoning. Understand why a model changes its settings, expresses uncertainty, and moves numbers through its layers.